



Uveal melanoma

Fractionated stereotactic radiotherapy for uveal melanoma, late clinical results

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ABSTRACT

Purpose: To determine local control, late toxicity and metastatic free survival (MFS) of patients treated with fractionated stereotactic radiation therapy (fSRT) for uveal melanoma (UM).

Methods and materials: Between 1999 and 2007, 102 UM patients were included in a prospective study of a single institution (median follow-up (FU) 32 months; median tumor thickness 6 mm); five fractions of 10 Gy were given. Primary endpoints were local tumor control and late toxicity (including visual outcome and eye preservation). Secondary endpoint was MFS.

Results: Local tumor control was achieved in 96% of the patients. Fifteen enucleations were performed, 2–85 months after radiation. Four eyes were enucleated because of local tumor progression. Nine patients developed grade 3 or 4 neovascular glaucoma (NVG), 19 developed severe retinopathy, 13 developed optic neuropathy grade 3 or 4, 10 developed cataract grade 3, and 10 patients suffered from keratitis sicca. Best corrected visual acuity (BCVA) decreased from a mean of 0.26 at diagnosis to 0.16, 3 months after radiation and it gradually declined to 0.03, 4 years after therapy. The 5-year actuarial MFS was 75% (95% CIs: 62–84%).

Conclusions: fSRT is an effective treatment modality for uveal melanoma with a good local control. With that, fSRT is a serious eye sparing treatment modality. However, our FU is relatively short. Also, the number of secondary enucleations is substantial, mainly caused by NVG.

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Uveal melanoma (UM) is the most common primary eye cancer in adults [1]. Still, UM occurs infrequently, having an annual age adjusted incidence of 4–7 per million [1]. Generally, uveal melanomas arise from the choroid (72%); however, they may also appear in the ciliary body (23%) or iris (5%). Currently next to enucleation, patients can be offered several eye sparing treatment options. The preferred treatment is influenced by a number of patient and tumor related factors, such as general condition, age, size and location of the tumor, visual acuity of the contralateral eye and the presence of metastases at diagnosis. Beside radiotherapy, transscleral resection, transretinal resection and phototherapy are used for the primary treatment of uveal melanomas. It is beyond the scope of this paper, however, to discuss these modalities.

Regarding radiotherapy, brachytherapy appears to be an effective treatment modality for UM [2–9]. For medium and large sized tumors, heavy particle radiation techniques (proton and helium ion beam) as well as stereotactic radiation therapy (SRT) are other eye sparing treatment options [10–14]. Single fraction SRT (“radi-

osurgery”) for UM, has been introduced in 1987, with local control rates between 90% and 98% [15,16]. The most important shortcoming of radiosurgery is the loss of radiobiological sparing of adjacent critical structures. Fractionated stereotactic radiotherapy (fSRT) reduces the effective dose in the surrounding tissues with a delivery of an equivalent effective dose to the tumor. In 1999, a longitudinal cohort study has been initiated in our institute, in which patients with UM were treated with fSRT. Preliminary results demonstrated that fSRT is well tolerated and can be safely implemented in the treatment of UM with excellent short term local control and mild acute side effects [17,18]. However, to determine the efficacy of fSRT, not only local control and survival at long-term, but also late toxicity is an issue to be studied in great detail. In this paper, the primary endpoints local tumor control and late toxicity, including visual outcome and eye preservation, as well as the secondary endpoint, metastatic free survival (MFS) of 102 UM patients are analyzed.

Methods and materials

Study design

In 1999, a prospective single center clinical trial was initiated in order to study both the safety and efficacy of fSRT for UM. In this

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